A Comparative Analysis of Machine Learning Approaches for DNA Sequence Classification Across Human, Chimpanzee, and Dog Genomes

Abstract

DNA sequence classification is an important task in bioinformatics, enabling the identification of genetic patterns and supporting the analysis of functional similarities across biological sequences. This study presents a machine learning-based approach for classifying DNA sequences into predefined gene classes using Human, Chimpanzee, and Dog datasets. The proposed methodology involves DNA sequence preprocessing, overlapping six-mer extraction, text-based feature representation, model training, and performance evaluation. The extracted six-mers are converted into space-separated text representations and transformed into numerical features using CountVectorizer with a four-token n-gram configuration. A Multinomial Naive Bayes classifier with a smoothing parameter of 0.1 is trained on the Human dataset using an 80:20 training–testing split. The trained model is evaluated on the held-out Human test set and applied to the Chimpanzee and Dog datasets using the same feature vocabulary and classifier. The experimental results show accuracies of 98.4% for Human, 99.3% for Chimpanzee, and 92.6% for Dog DNA sequences. The corresponding weighted F1-scores are 98.4%, 99.3%, and 92.5%, respectively, while the reported macro-averaged ROC-AUC scores are 0.998, 0.999, and 0.983. These findings indicate that k-mer-based text representation combined with Multinomial Naive Bayes can provide a strong baseline for DNA sequence classification. However, the cross-dataset results should be interpreted cautiously because the classifier is trained only on Human data, and the evaluation datasets may differ in their sequence and class distributions. Furthermore, the feature extraction strategy and multiclass ROC-AUC calculations require careful validation. This study establishes a baseline for further investigation of DNA sequence classification using alternative sequence representations, comparative machine learning models, and consistent cross-dataset evaluation protocols.

Introduction

DNA sequence classification is an important task in bioinformatics, with applications in gene identification, genomic annotation, evolutionary analysis, and the study of genetic similarities across species [1], [2]. DNA sequences contain biological information encoded through nucleotide patterns, and identifying meaningful patterns within these sequences can help researchers investigate gene functions and relationships between organisms. However, the large number of possible nucleotide combinations and the variability of genomic sequences make accurate and computationally efficient classification a challenging problem.

Machine learning has emerged as a useful approach for DNA sequence classification because it can identify discriminative patterns from sequence-derived features without requiring manually designed biological rules for every classification task [1], [2]. Among different feature extraction techniques, the *k*-mer representation is widely used in computational genomics. It divides a DNA sequence into overlapping subsequences of length *k* and represents each sequence through the occurrence or frequency of these subsequences [3]. This representation converts biological sequences into numerical features suitable for conventional machine learning algorithms. In particular, *k*-mer-based bag-of-words representations can capture local nucleotide patterns and provide a practical approach for classifying DNA sequences.

Among conventional machine learning algorithms, Multinomial Naive Bayes (MNB) is a computationally efficient probabilistic classifier that is particularly suitable for discrete count-based features. Its relatively simple training procedure and ability to handle high-dimensional sparse representations make it an attractive baseline for text-like DNA sequence classification [2], [4]. By combining *k*-mer feature extraction with MNB, DNA sequences can be transformed into numerical representations and assigned to predefined gene-family classes. Nevertheless, classification performance may vary across species because genetic differences can influence the distribution of nucleotide patterns and the transferability of learned features.

Although *k*-mer-based classification and machine learning have been investigated in bioinformatics, evaluating how a classifier trained on one species performs on

sequences from other species remains an important experimental consideration [2], [3]. Comparisons involving human, chimpanzee, and dog DNA can provide insight into the extent to which learned sequence patterns generalize across organisms. However, reliable conclusions require a consistent feature-extraction pipeline, appropriate data partitioning, and clearly distinguished evaluation protocols. In particular, performance measured on a held-out test set from the training species should not be directly treated as equivalent to performance on complete datasets from other species. These considerations motivate a comparative evaluation that examines both within-species classification and cross-species transfer using a consistent machine learning approach.

To address this objective, this study investigates *k*-mer-based DNA sequence classification using Multinomial Naive Bayes on human, chimpanzee, and dog datasets. DNA sequences are represented using overlapping six-nucleotide subsequences, which are converted into a sparse count-based feature representation for classification. The MNB classifier is trained using human DNA sequences and evaluated on a held-out human test set. Its transfer performance is then examined using the chimpanzee and dog datasets. Model performance is assessed using accuracy, weighted precision, weighted recall, weighted F1-score, confusion matrices, and receiver operating characteristic (ROC) analysis with area under the curve (AUC). This evaluation provides a comparative view of classification performance across the three species under a shared feature vocabulary and classification model.

The main contributions of this study are summarized as follows:

1. **A *k*-mer-based DNA classification pipeline:** A feature-based framework is developed using overlapping six-nucleotide subsequences and count-based vectorization for Multinomial Naive Bayes classification.
2. **Within-species performance evaluation:** The classifier is evaluated on a held-out human DNA test set using multiple classification metrics and a confusion matrix.
3. **Cross-species transfer analysis:** The trained human DNA classifier is applied to chimpanzee and dog sequences to investigate how well the learned gene-family classification patterns transfer across species.

4. **Comparative performance assessment:** Accuracy, weighted precision, weighted recall, weighted F1-score, and ROC-AUC are used to compare the observed results across the three datasets and identify differences in classification performance.

Related Work

DNA sequence classification has become an important research area in bioinformatics because it supports gene-family identification, genomic annotation, and comparative genomic analysis. Traditional approaches often rely on sequence alignment and biological similarity measures, whereas machine learning methods learn discriminative patterns from numerical representations of nucleotide sequences. Among these methods, k-mer-based feature extraction has gained attention because it converts DNA sequences into fixed-dimensional representations suitable for conventional classification algorithms.

Several studies have examined k-mer methods and their applications in bioinformatics. Moeckel et al. [1] reviewed k-mer-based approaches and highlighted their usefulness in diverse sequence-analysis tasks. A k-mer represents a contiguous subsequence of length k, and overlapping k-mers capture local nucleotide patterns that may distinguish different biological sequence classes. Similarly, Yang et al. [2] reviewed machine learning algorithms for DNA sequence mining and discussed the importance of sequence representation, feature extraction, and classifier selection in classification performance. These studies provide a methodological foundation for transforming DNA sequences into numerical features for supervised learning.

Conventional machine learning algorithms have also been investigated for DNA sequence classification. Juneja et al. [3] explored machine learning approaches to DNA sequence classification, demonstrating the relevance of supervised learning for identifying patterns in biological sequences. Among these approaches, Multinomial Naive Bayes (MNB) is particularly suitable for count-based features because it

estimates class probabilities from feature frequencies. When combined with a bag-of-words representation, MNB provides a computationally efficient baseline for evaluating whether sequence-derived patterns can distinguish multiple gene-related classes. However, its effectiveness depends on the selected feature representation, the distribution of training samples, and the generalization of learned patterns to unseen sequences.

Cross-species classification introduces additional challenges because nucleotide patterns and class distributions may vary between organisms. Patil et al. [4] investigated k-mer-based machine learning for predicting DNA sequence similarity across species, including human, chimpanzee, and dog sequences. Their work demonstrates the relevance of sequence-derived features and conventional classifiers to comparative genomic analysis. Although these organisms share evolutionary relationships, differences in nucleotide composition and gene-family characteristics can influence classification performance across species. Consequently, evaluating a model on sequences from multiple organisms can provide insights into its transferability beyond the species used for training.

Despite these advances, comparisons of DNA sequence classification performance require consistent feature extraction and rigorous evaluation protocols. In particular, evaluation on the complete dataset of a different species should be distinguished from evaluation on an independent held-out test set. Otherwise, reported performance may not provide a directly comparable estimate of generalization. In addition, the use of higher-order n-grams over overlapping k-mer tokens can capture local sequence patterns, but it also creates a high-dimensional feature space that may increase computational requirements. To examine these considerations, the present study applies six-nucleotide k-mer tokenization, four-token n-gram feature extraction, and a Multinomial Naive Bayes classifier to human DNA sequences, followed by cross-species evaluation on chimpanzee and dog datasets. The experimental analysis compares accuracy, precision, recall, F1-score, and ROC-AUC to characterize classification performance across the three organisms.

Methodology

This study investigates DNA sequence classification using a conventional machine learning pipeline based on k-mer tokenization, n-gram feature extraction, and a Multinomial Naive Bayes (MNB) classifier. The experimental framework evaluates the classification of gene-related categories in human DNA sequences and examines the transferability of the trained classifier to chimpanzee and dog datasets. The methodology consists of dataset preparation, DNA sequence representation, feature extraction, model training, cross-species evaluation, and performance assessment.

A. Dataset Description and Preparation

The experiments utilize the DNA Sequence Dataset available on Kaggle, comprising three tabular datasets: human.txt, chimpanzee.txt, and dog.txt. Each dataset contains DNA sequences and their corresponding class labels, representing seven gene-related categories indexed from 0 to 6. When loaded using the original tabular format, the datasets contain 4,380 human sequences, 1,682 chimpanzee sequences, and 820 dog sequences, respectively. An additional FASTA file, example_dna.fa, is examined during preliminary data exploration but is excluded from the classification experiments. Each record consists of a nucleotide sequence and its associated class label. The sequence and label columns are separated to establish the input data and target variables. The nucleotide sequences are subsequently transformed into text-based representations suitable for machine learning. The same preprocessing and feature extraction procedures are applied to all three species to maintain a consistent representation.

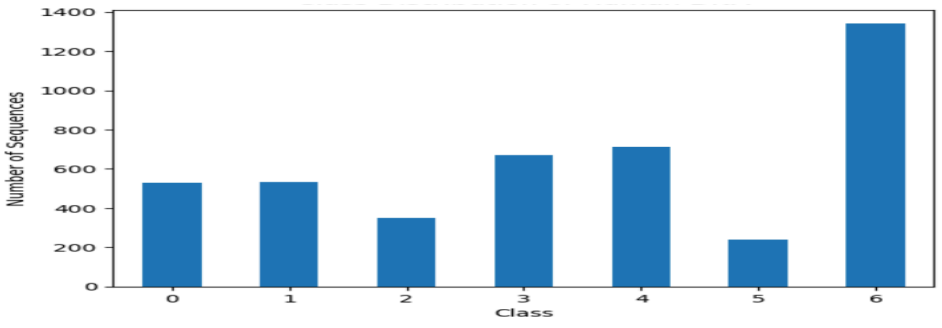


Figure 1. Class Distribution of Human DNA.

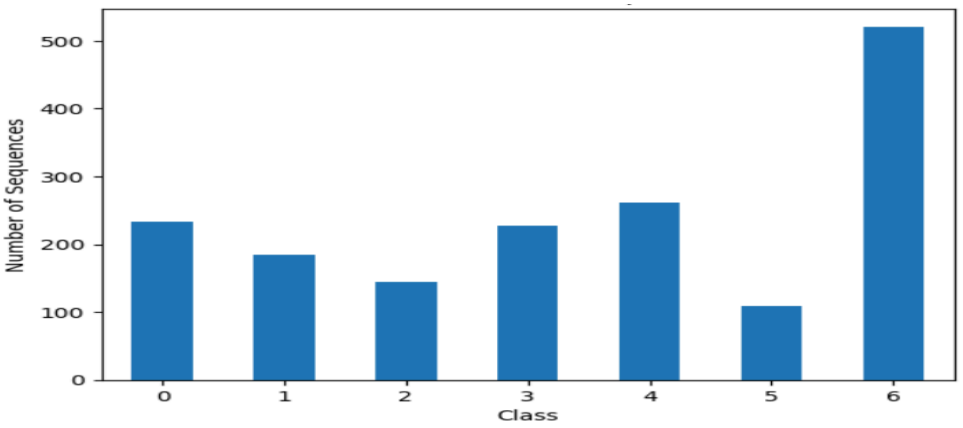


Figure 2. Class Distribution of Chimpanzee DNA.

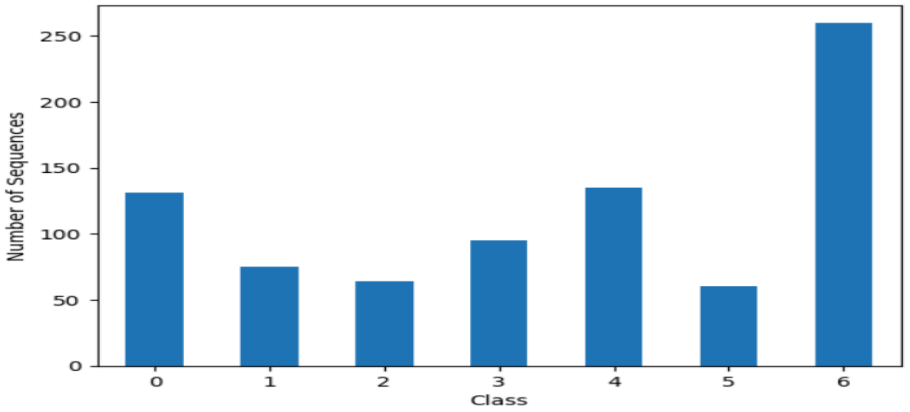


Figure 3. Class Distribution of Dog DNA.

B. K-mer-Based DNA Sequence Representation

DNA sequences are represented using overlapping six-nucleotide subsequences, commonly known as 6-mers. Each sequence is processed using a sliding window of six nucleotides, moving forward by one nucleotide at a time. This approach generates overlapping segments that preserve local nucleotide patterns.

For example, the DNA sequence ATGCCCA produces two 6-mer tokens: ATGCCC and TGCCCA. The extracted tokens are converted into lowercase strings and concatenated using whitespace separators. This transformation converts biological sequences into a token-based representation that can be processed using standard text vectorization techniques.

The same six-mer tokenization procedure is applied to human, chimpanzee, and dog DNA sequences to ensure consistency across the datasets.

C. N-gram Feature Extraction

The tokenized DNA sequences are converted into numerical feature vectors using the CountVectorizer implementation from Scikit-learn. The vectorizer is configured with ngram_range=(4,4) to extract groups of exactly four consecutive 6-mer tokens.

Because adjacent 6-mers overlap by five nucleotides, four consecutive tokens represent a local sequence pattern spanning nine nucleotide positions. The extracted features therefore represent four-token n-grams over overlapping six-nucleotide segments rather than conventional four-nucleotide k-mers.

The vectorizer constructs a vocabulary of observed n-gram features and represents each sequence through the occurrence counts of these features. In the current implementation, the human dataset produces a feature matrix containing 4,380 samples and 232,414 features. The fitted vocabulary is then used to transform chimpanzee and dog sequences into the same feature space.

For a leakage-controlled experimental design, the vectorizer should be fitted exclusively on the human training subset after the train–test split. The resulting vocabulary should then be applied unchanged to the human test set and the cross-species datasets.

D. Classification Model and Training Procedure

The Multinomial Naive Bayes classifier is selected because it supports non-negative, count-based features and provides computationally efficient probabilistic classification. The classifier estimates class probabilities based on the observed frequencies of the extracted n-gram features, assuming conditional independence between features given the class.

Additive smoothing is applied using a parameter value of 0.1 to reduce the influence of features that are absent from a particular class during training.

The human dataset is divided into training and testing subsets using an 80:20 split, with a fixed random state of 42 to support reproducibility. This partition produces 3,504 training samples and 876 testing samples. The classifier is trained on the human training data and corresponding class labels. After training, the model predicts the class labels of the held-out human test samples.

The same trained classifier is subsequently applied to the chimpanzee and dog datasets to investigate its ability to recognize gene-related categories across different species.

E. Cross-Species Evaluation Strategy

Cross-species evaluation is conducted by applying the human-trained classifier to chimpanzee and dog DNA sequences. The predicted class labels are compared with the available reference labels to assess classification performance across species.

This approach provides an initial assessment of whether patterns learned from human DNA sequences can be transferred to other organisms. However, the current notebook evaluates the complete chimpanzee and dog datasets rather than independent held-out subsets. Therefore, these measurements are described as full-dataset cross-species evaluation results rather than independent test-set estimates.

For a more rigorous assessment, separate evaluation subsets should be established for chimpanzee and dog sequences. Duplicate or highly similar sequences should also be examined to reduce potential evaluation bias.

F. Performance Evaluation Metrics

The classifier is evaluated using accuracy, weighted precision, weighted recall, weighted F1-score, confusion matrices, and the area under the receiver operating characteristic curve (ROC-AUC). Accuracy measures the proportion of correctly classified samples. Precision indicates how many predicted class assignments are correct, while recall measures how effectively the classifier identifies samples belonging to each actual class. The F1-score combines precision and recall into a single measure. Weighted precision, recall, and F1-score account for differences in

the number of samples across classes. Confusion matrices are used to examine class-specific prediction errors and identify categories that are frequently misclassified. Macro ROC-AUC is calculated using a one-vs-rest strategy to assess discrimination across the seven classes. ROC-AUC should be calculated from predicted class probabilities or decision scores on clearly defined evaluation subsets.

G. Experimental Workflow

The experimental workflow comprises dataset preparation, six-mer tokenization, four-token n-gram feature extraction, human-based model training, cross-species prediction, and performance evaluation. The Multinomial Naive Bayes classifier is trained on human DNA sequences and evaluated on the held-out human test set. The trained model is then applied to chimpanzee and dog sequences using the same feature representation. The current notebook reports an accuracy of 98.4% on the held-out human test set, 99.3% on the complete chimpanzee dataset, and 92.6% on the complete dog dataset. These values represent the existing experimental outputs and should not be interpreted as directly comparable independent test-set results because the evaluation procedures differ. The final results should be reassessed after applying training-only vocabulary fitting and a consistent, independent evaluation protocol.

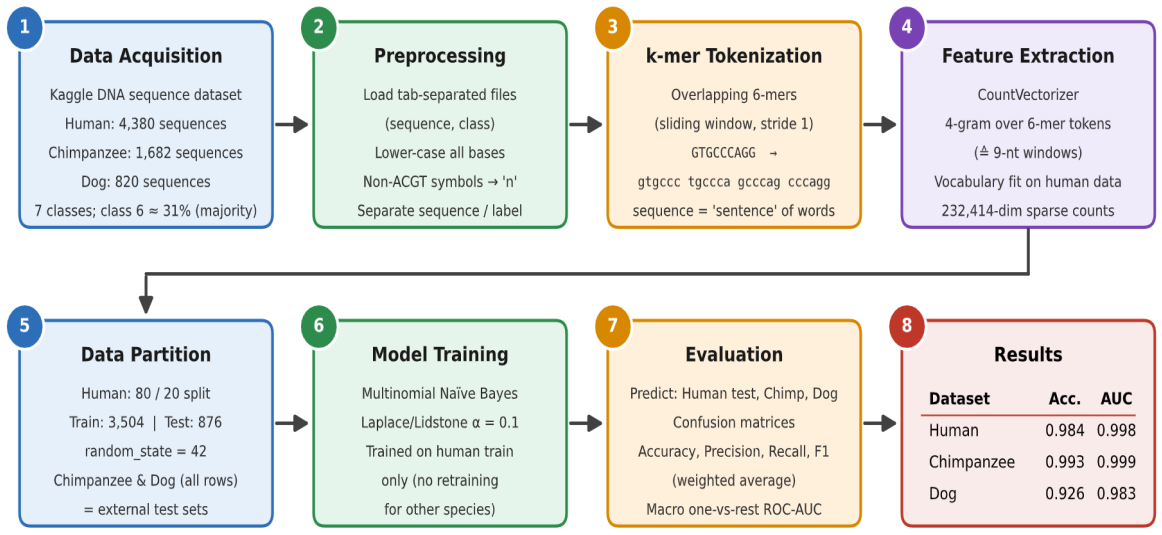


Figure 4. Methodological Pipeline for DNA Sequence Data Processing and Classification

Experimental Setup

This section describes the experimental environment, dataset, preprocessing procedures, feature extraction method, and evaluation protocol used to assess DNA sequence classification using a Multinomial Naive Bayes classifier. Experiments were conducted using human, chimpanzee, and dog DNA sequences from the Kaggle DNA Sequence Dataset.

A. Experimental Platform

All experiments were implemented in a Kaggle Notebook environment using Python. Pandas and NumPy were used for data processing, while Scikit-learn supported feature extraction, model training, and evaluation. Biopython was used to inspect the supplementary FASTA file, and Matplotlib and Seaborn were employed to visualize class distributions, confusion matrices, and classification performance.

Table I. Experimental Software Environment

Category	Tool/Library	Purpose
Platform	Kaggle Notebook	Experimental environment
Programming Language	Python	Version to be recorded
Data processing	Pandas and NumPy	Dataset loading and processing
Sequence processing	Biopython	FASTA parsing
Feature extraction	Scikit-learn CountVectorizer	Four-token n-gram counts

Classification	Multinomial Naive Bayes	$\alpha = 0.1$
Visualization	Matplotlib and Seaborn	Plots and confusion matrices

B. Dataset

The experiments used the DNA Sequence Dataset available on Kaggle. The dataset contains DNA sequences from three organisms: human, chimpanzee, and dog. Each sequence is associated with one of seven gene-related classes, represented by labels from 0 to 6. The main classification pipeline used the human.txt, chimpanzee.txt, and dog.txt files. The supplementary example_dna.fa file was inspected but was not used in model training or evaluation.

Table II. Dataset Statistics

Dataset	**Number of sequences**	**Features**
Human	4,380	DNA sequence and class label
Chimpanzee	1,682	DNA sequence and class label
Dog	820	DNA sequence and class label
Total	6,882	Seven gene-related classes

The dataset sizes were obtained using pandas.read_table(). Class distributions were examined through bar charts to understand the representation of different gene-related categories.

C. Feature Extraction and Model Configuration

DNA sequences were represented using an overlapping six-mer approach. A sliding window of six nucleotides was moved across each sequence to extract consecutive k-mers. The extracted six-mers were joined into space-separated text documents and converted into numerical features using Scikit-learn's CountVectorizer with ngram_range=(4,4).

Thus, each text feature represents a pattern of four consecutive six-mer tokens, spanning nine nucleotide positions. The vectorizer was fitted on the human DNA sequences, generating a vocabulary of 232,414 features. The same vocabulary was used to transform the chimpanzee and dog sequences, ensuring consistent feature dimensions across datasets.

Table III. Feature Extraction Configuration

Parameter	**Setting**
Sequence representation	Overlapping k-mers
K-mer length	6 nucleotides
Text vectorization	CountVectorizer
N-gram range	(4, 4)
Human feature matrix	4380×232414
Chimpanzee feature matrix	1682×232414
Dog feature matrix	820×232414

D. Evaluation Protocol

The human dataset was divided into training and testing subsets using an 80:20 ratio, with random_state=42. This resulted in 3,504 training samples and 876 testing samples. The Multinomial Naive Bayes classifier was trained on the human training subset and evaluated on the held-out human test set. The trained model was also applied to the complete chimpanzee and dog datasets to investigate its performance across species. Performance was measured using accuracy, weighted precision, weighted recall, weighted F1-score, and macro-averaged ROC-AUC. Confusion matrices were generated to examine class-level prediction errors.

Results Analysis and Discussion

This section presents a comparative evaluation of a Multinomial Naive Bayes (MNB) classifier for DNA sequence classification using human, chimpanzee, and dog DNA datasets. The experimental pipeline converts DNA sequences into overlapping six-nucleotide k-mers and subsequently extracts four-gram features using the CountVectorizer method. The resulting feature vectors are classified into seven gene-related categories, labeled from 0 to 6. The classifier is trained on the human DNA dataset and evaluated using a held-out human test set and the available chimpanzee and dog datasets.

The analysis is organized into five parts: classification performance, cross-species evaluation, ROC-AUC analysis, confusion matrix analysis, and overall model limitations and implications. The objective is to investigate the effectiveness of k-mer-based text representation for DNA classification and assess how well a classifier trained on human sequences performs on sequences from other species.

A. Classification Performance Evaluation

Classification performance is evaluated using accuracy, weighted precision, weighted recall, and weighted F1-score. Accuracy measures the proportion of

correctly classified sequences, while precision, recall, and F1-score provide complementary perspectives on prediction quality across the seven classes.

Dataset	Accuracy	Weighted precision	Weighted Recall	Weighted F1-Score
Human (test set)	0.984	0.984	0.984	0.984
Chimpanzee	0.993	0.994	0.993	0.993
Dog	0.926	0.934	0.926	0.925

Table I. Reported classification performance of the Multinomial Naive Bayes model.

The human test set achieved an accuracy of 98.4%, with weighted precision, recall, and F1-score also reaching 98.4%. These results indicate that the classifier can distinguish the gene-related categories effectively within the evaluated human dataset. The high F1-score suggests that the model's classification performance remains strong when considering both precision and recall.

The chimpanzee evaluation produced the highest reported accuracy of 99.3%, together with a weighted precision of 99.4% and an F1-score of 99.3%. This result suggests that the learned k-mer patterns correspond well to the class distributions in the chimpanzee data. However, this score should be interpreted cautiously because the chimpanzee dataset was evaluated in its entirety rather than on an independent held-out subset.

In contrast, the dog dataset achieved an accuracy of 92.6% and a weighted F1-score of 92.5%. Although the performance remains relatively high, it is lower than the reported human and chimpanzee results. Differences in sequence composition, the distribution of gene-related classes, and the degree of similarity between the species' sequences may contribute to this variation. The present experiment, however, does not independently establish which factor is responsible.

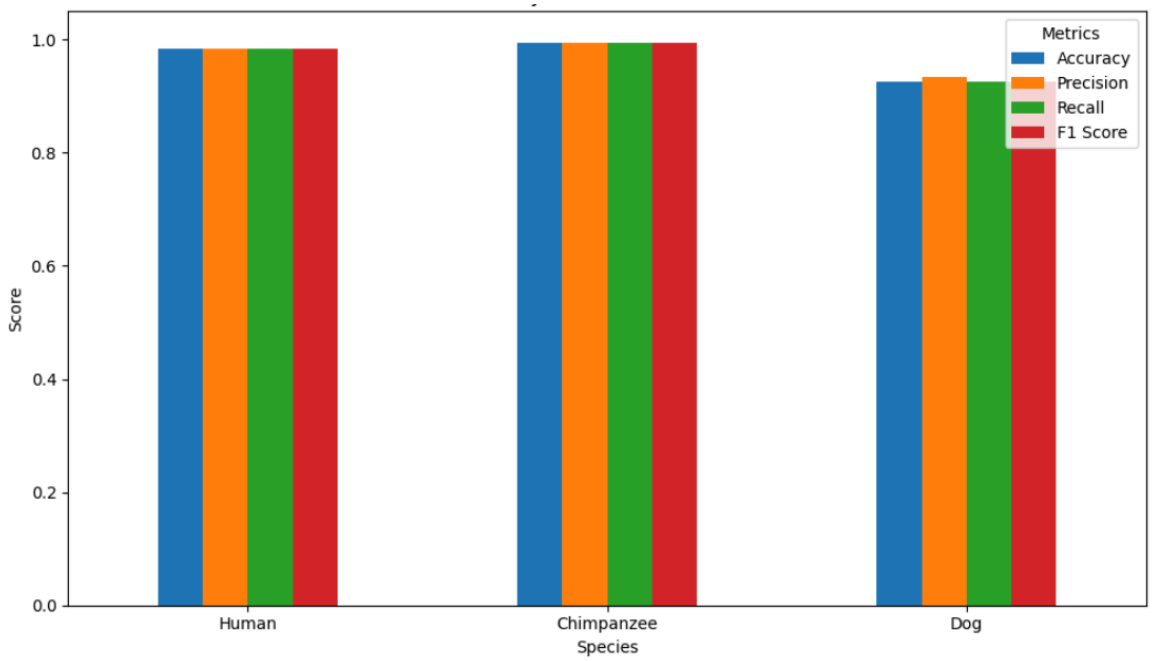


Figure 5. Comparative classification performance across the human, chimpanzee, and dog datasets using accuracy, weighted precision, weighted recall, and weighted F1-score.

B. Cross-Species Classification Analysis

An important objective of this experiment is to investigate whether a model trained on human DNA sequences can classify gene-related categories in DNA sequences from other species. The feature extraction pipeline generates a shared vocabulary of 232,414 four-gram features from overlapping six-mer tokens. Chimpanzee and dog sequences are then transformed using this same vocabulary before prediction. The human dataset contains 4,380 sequences, of which 3,504 are used for training and 876 for testing. The chimpanzee and dog datasets contain 1,682 and 820 sequences, respectively. The MNB classifier is trained only on the human training partition and then applied to the other species.

The results show different levels of cross-species performance. Chimpanzee sequences achieve a reported accuracy of 99.3%, whereas dog sequences achieve 92.6%. The stronger chimpanzee result is consistent with the possibility that the model's learned sequence patterns are more transferable to the chimpanzee dataset than to the dog dataset. Nevertheless, the observed difference cannot by itself

demonstrate a biological cause or establish generalization to unseen populations. A critical methodological distinction is that the human result is calculated on a held-out test set, while the chimpanzee and dog metrics are calculated over their complete datasets. Therefore, the three scores do not represent identical evaluation protocols. The cross-species results should be treated as preliminary evidence of transfer performance rather than a definitive comparison of generalization ability.

C. ROC-AUC Analysis

In addition to discrete classification metrics, the experiment evaluates the model using the macro-averaged area under the receiver operating characteristic curve (ROC-AUC). ROC-AUC summarizes the model's ability to distinguish positive from negative examples across different decision thresholds. Macro averaging gives equal importance to the AUC calculated for each class.

Dataset	**Reported Macro ROC-AUC**
Human (test set)	0.998
Chimpanzee	0.999
Dog	0.983

Table II. Reported macro ROC-AUC values.

The human dataset achieves a reported macro ROC-AUC of 0.998, indicating strong class-separation performance on the held-out human test set. The chimpanzee dataset records 0.999, while the dog dataset records 0.983. All three reported values are high, although the dog result is comparatively lower. These results complement the accuracy and F1-score findings by evaluating the classifier's probability-based discrimination across classes. However, a high ROC-AUC does not necessarily imply perfect classification at the default prediction threshold, nor does it establish reliable performance on independently sampled data.

The ROC-AUC implementation should also be verified before publication. The reported macro calculation uses label_binarize with seven classes, but scikit-learn

may represent a seven-class target using a reduced binary indicator matrix in some configurations. The code's assumption that the binarized matrix has seven columns should be checked explicitly. The macro-AUC and micro-averaged ROC curves should be regenerated using a validated multiclass evaluation procedure, with the same class ordering used for predict_proba.

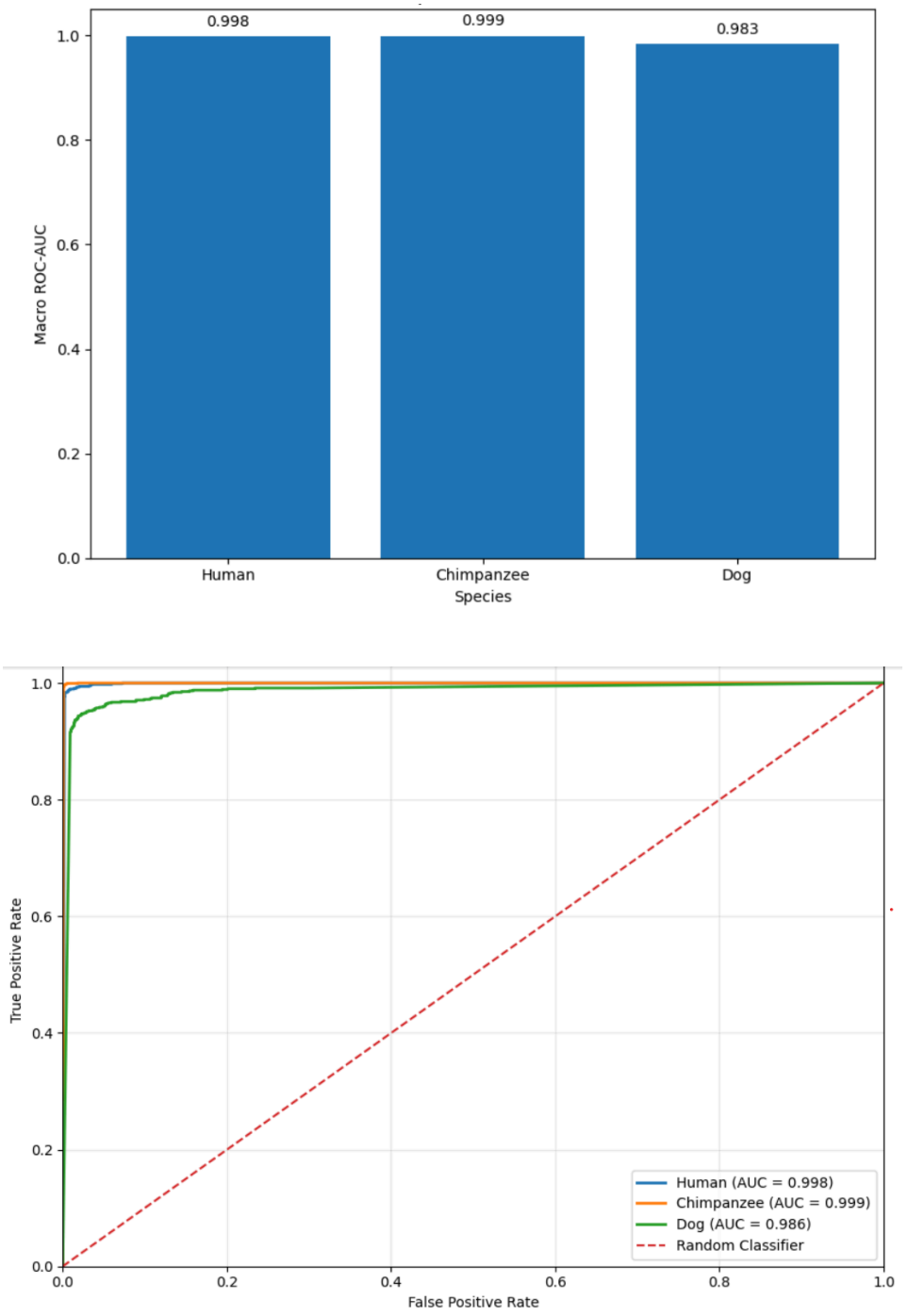


Figure 6. Reported macro ROC-AUC comparison for the human, chimpanzee, and dog datasets.

D. Confusion Matrix Analysis

The confusion matrices provide a class-level view of correct and incorrect predictions. In the human test set, the diagonal entries indicate correct predictions, while off-diagonal entries represent misclassifications. The matrix shows strong classification performance across all seven classes, with most errors concentrated in a small number of categories.

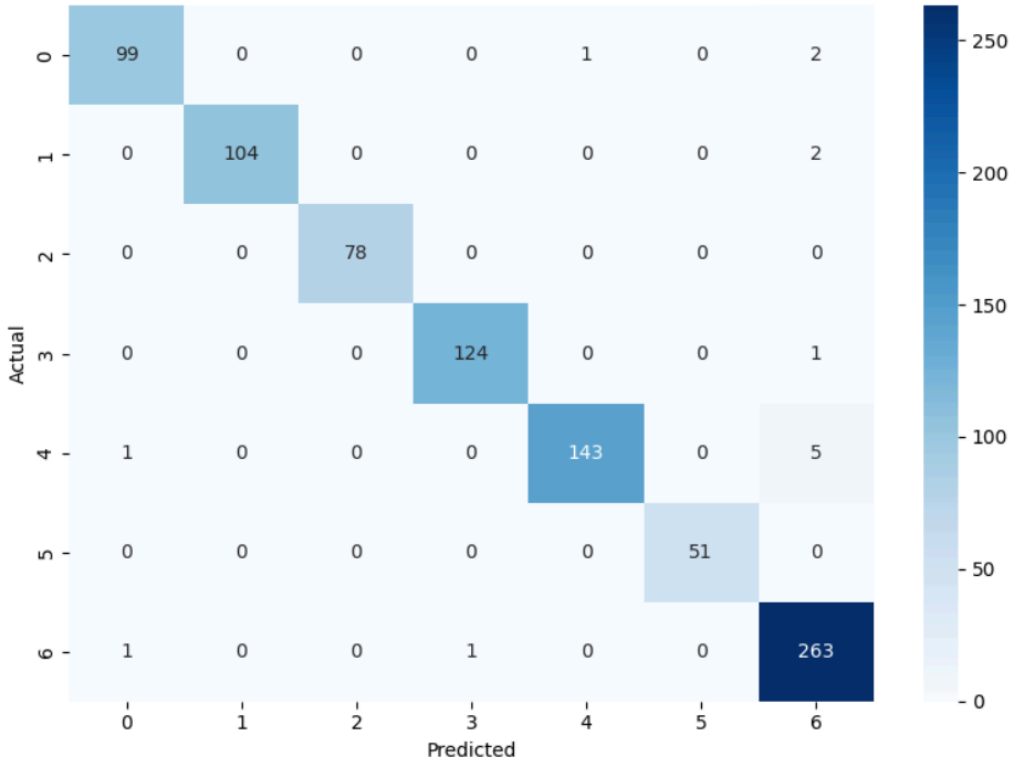


Figure 7. Confusion matrices for human DNA classification.

human class 5 has 51 correct predictions and no off-diagonal errors in the reported matrix. Class 6 has 263 correct predictions, with a small number of samples assigned to other classes. These patterns suggest that the learned k-mer features distinguish many of the evaluated categories effectively, although some classes exhibit limited overlap in their feature representations.

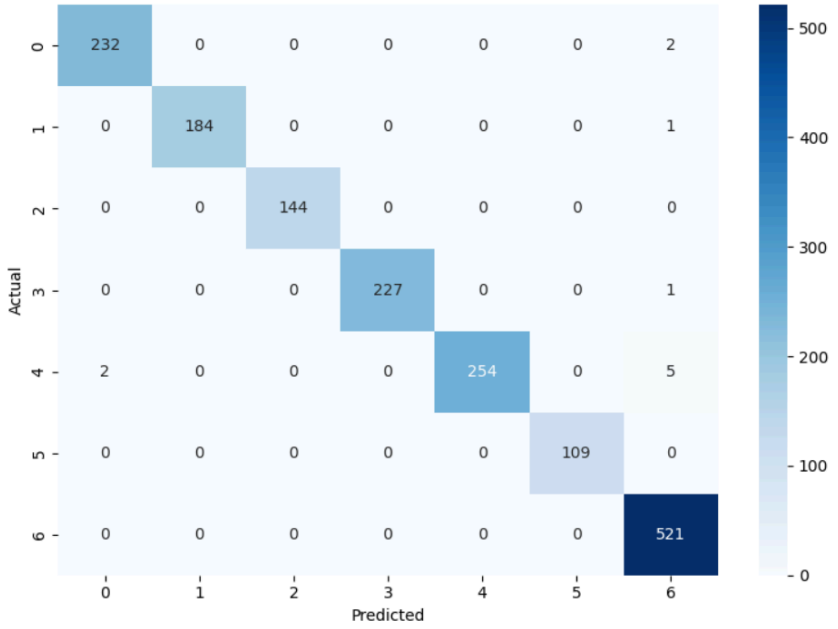


Figure 8. Confusion matrices for Chimpanzee DNA classification.

The chimpanzee confusion matrix similarly reports a large concentration of predictions along the diagonal. However, its matrix is calculated on the complete chimpanzee dataset, so it should not be interpreted as an independent test of the classifier's generalization.

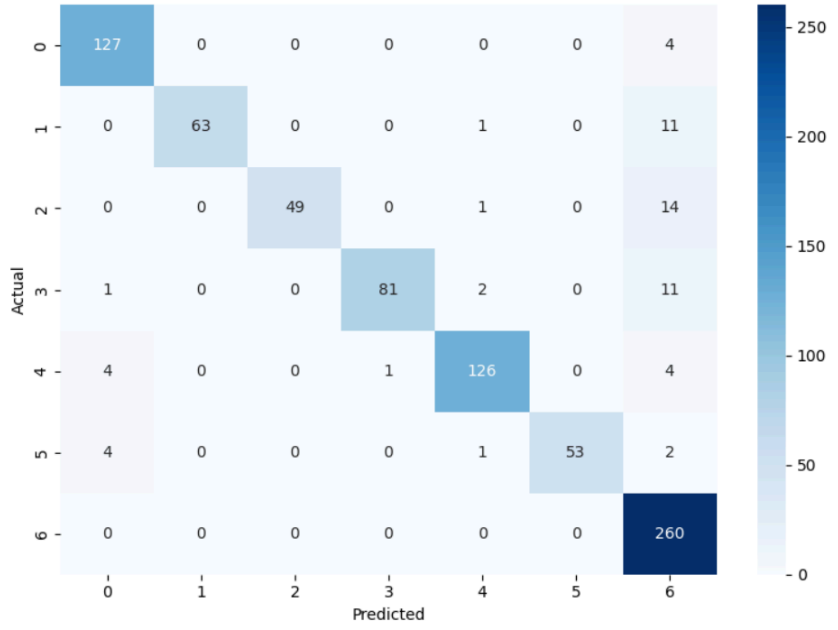


Figure 9. Confusion matrices for Dog DNA classification.

For dog DNA, the matrix shows more off-diagonal predictions. In particular, several sequences from classes 1, 2, 3, and 5 are assigned to class 6. This pattern suggests that the current representation and classifier may have difficulty separating certain dog sequence categories. The observed errors could reflect differences in class-specific sequence patterns, feature coverage, or class overlap; further class-level analysis is required to determine the underlying reasons.

E. Discussion and Limitations

The experimental results demonstrate that overlapping k-mer extraction combined with four-gram CountVectorizer features and Multinomial Naive Bayes can produce high reported classification scores on the evaluated DNA datasets. The approach converts biological sequences into sparse, count-based representations that can be processed efficiently by a conventional machine-learning classifier. Its relatively simple architecture also provides a useful baseline for comparison with more computationally demanding sequence-classification methods.

The human test-set accuracy of 98.4% and the reported chimpanzee accuracy of 99.3% indicate strong performance under the current experimental procedure. The dog accuracy of 92.6%, however, suggests that classification performance varies across datasets. The ROC-AUC results show a similar pattern, with reported values ranging from 0.983 to 0.999.

Several limitations must be addressed before these results can support strong research conclusions. First, the CountVectorizer vocabulary is fitted on the complete human dataset before the train-test split. Although this does not use the test labels, it exposes feature-selection information from the test sequences and introduces a potential source of data leakage. The vectorizer should instead be fitted exclusively on the training partition and then applied to the human test set and cross-species data.

Second, the chimpanzee and dog metrics are computed on the same complete datasets used for evaluation, without independent holdout subsets. A rigorous cross-species experiment should reserve independent test samples for each species and report the evaluation protocol clearly. Third, the exploratory ordinal and one-hot

encoding functions are not part of the main classification pipeline; the reported classifier uses six-mer text tokens and four-gram count features. These encoding methods should therefore not be described as model inputs in the final paper.

Finally, the implementation should verify class ordering and multiclass ROC-AUC calculations, and should report class-wise precision, recall, F1-score, and support where possible. Repeating the experiments with stratified splits and multiple random seeds would provide a more reliable estimate of performance variability.

Overall, the findings support k-mer-based feature extraction with Multinomial Naive Bayes as a promising baseline for gene-related DNA sequence classification. However, the reported scores should be considered preliminary, particularly for cross-species evaluation, until the feature extraction and evaluation procedures are corrected and the results are reproduced under leakage-controlled experimental conditions.

Conclusion

This study presented a comparative evaluation of Multinomial Naive Bayes for DNA sequence classification using human, chimpanzee, and dog genomic sequence datasets. The proposed workflow transformed DNA sequences into overlapping six-nucleotide k-mers and subsequently extracted four-gram features using the CountVectorizer method. A shared feature vocabulary containing 232,414 features was constructed from the human DNA sequences. The classifier was trained using 80% of the human dataset, with the remaining 20% reserved for testing, and the trained model was subsequently evaluated on the chimpanzee and dog datasets to investigate its performance across different species.

The experimental results demonstrated strong classification performance on the human test set, achieving an accuracy of 98.4%, weighted precision of 98.4%, weighted recall of 98.4%, and weighted F1-score of 98.4%. The chimpanzee dataset produced an accuracy of 99.3% and an F1-score of 99.3%, while the dog dataset

achieved an accuracy of 92.6% and an F1-score of 92.5%. The reported macro ROC-AUC values were 0.998 for Human, 0.999 for chimpanzee, and 0.983 for dog sequences. These results indicate that the six-mer and four-gram feature representation, combined with Multinomial Naive Bayes, can capture informative sequence patterns associated with the seven gene-related classes. The confusion matrices also revealed that classification errors were more noticeable for dog sequences, particularly where samples from several classes were predicted as class 6.

Despite these promising results, the findings must be interpreted in light of important methodological limitations. The human dataset was split into training and test subsets after the feature vocabulary had been fitted to the complete human dataset, potentially introducing vocabulary-level data leakage. Furthermore, the chimpanzee and dog datasets were evaluated in their entirety rather than through independent held-out test sets. Consequently, their scores are not directly comparable to the human test-set results and should be considered preliminary cross-species evaluation results rather than definitive estimates of generalization performance. The supplementary ordinal and one-hot encoding procedures were exploratory preprocessing demonstrations and were not used in the final classifier.

Future work should fit the feature extraction pipeline exclusively on the training data, employ stratified train–validation–test splits, and evaluate each species using clearly defined independent test sets. Additional experiments could compare Multinomial Naive Bayes with Support Vector Machines, Logistic Regression, Random Forest, and other suitable sequence-classification methods. Class-wise precision, recall, and F1-scores, repeated experiments with different random seeds, and carefully validated multiclass ROC-AUC calculations would further strengthen the reliability of the findings.

In conclusion, this study demonstrates the potential of a computationally straightforward k-mer-based machine learning pipeline for classifying gene-related categories in DNA sequences. Although the initial results are encouraging, rigorous validation and leakage-free evaluation are necessary before the approach can be considered reliable for broader genomic classification applications.

References

[1] A. Yang et al., “Review on the application of machine learning algorithms in the sequence data mining of DNA,” *Frontiers in Bioengineering and Biotechnology*, vol. 8, Art. no. 1032, 2020, doi: 10.3389/fbioe.2020.01032.

[2] K. Vervier, P. Mahé, M. Tournoud, J.-B. Veyrieras, and J.-P. Vert, “Large-scale machine learning for metagenomics sequence classification,” *Bioinformatics*, vol. 32, no. 7, pp. 1023–1032, 2016, doi: 10.1093/bioinformatics/btv683.

[3] D. E. Wood and S. L. Salzberg, “Kraken: Ultrafast metagenomic sequence classification using exact alignments,” *Genome Biology*, vol. 15, Art. no. R46, 2014, doi: 10.1186/gb-2014-15-3-r46.

[4] H. Gunasekaran et al., “Analysis of DNA sequence classification using CNN and hybrid models,” *Computational and Mathematical Methods in Medicine*, vol. 2021, Art. no. 1835056, 2021, doi: 10.1155/2021/1835056.

[5] A. McCallum and K. Nigam, “A comparison of event models for Naive Bayes text classification,” in *Proceedings of the AAAI-98 Workshop on Learning for Text Categorization*, 1998, pp. 41–48.

[6] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge University Press, 2008.

[7] P. J. A. Cock et al., “Biopython: Freely available Python tools for computational molecular biology and bioinformatics,” *Bioinformatics*, vol. 25, no. 11, pp. 1422–1423, 2009, doi: 10.1093/bioinformatics/btp163.

[8] D. J. Lipman and W. R. Pearson, “Rapid and sensitive protein similarity searches,” *Science*, vol. 227, no. 4693, pp. 1435–1441, 1985, doi: 10.1126/science.2983426.

[9] P. A. Pearson and D. J. Lipman, “Improved tools for biological sequence comparison,” *Proceedings of the National Academy of Sciences*, vol. 85, no. 8, pp. 2444–2448, 1988, doi: 10.1073/pnas.85.8.2444.

[10] A. Kautsar, “DNA sequence classification using machine learning models based on k-mer features,” *Journal of Computers and Digital Business*, vol. 4, no. 2, pp. 100–105, 2025, doi: 10.56427/jcbd.v4i2.762.

[11] S. O. Ozan, “DNA sequence classification with compressors,” 2024. Publication details require verification before citation.

[12] J. R. Quinlan, *C4.5: Programs for Machine Learning*. San Mateo, CA, USA: Morgan Kaufmann, 1993.

[13] T. Fawcett, “An introduction to ROC analysis,” *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006, doi: 10.1016/j.patrec.2005.10.010.

[14] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[15] M. A. Hall, “Correlation-based feature selection for machine learning,” Ph.D. dissertation, Dept. Comput. Sci., Univ. Waikato, Hamilton, New Zealand, 1999.